From 191 to 25 covariates: a parsimonious linear model for IFC

Multicollinearity pruning, significance filtering, and the geographic-hierarchy recovery

Applied Statistics Project — Extension Report

Abstract

This extension report documents an iteration of the IFC linear benchmark made in response to instructor feedback. The original LASSO benchmark (191 covariates, $R^2 = 0.825$) was considered too dense to discuss substantively in a poster setting. We therefore derived a *parsimonious* model in two steps. First, we eliminated multicollinearity by iteratively dropping the covariate with the highest Variance Inflation Factor until no VIF exceeded 10 (191 $\to$ 157 covariates). Second, we ranked the survivors by standardised effect, kept the top 25, and applied a post-hoc significance filter that replaces any non-significant predictor with the next candidate from the ranking. The resulting 25-covariate model reaches $R^2 = 0.784$ on the same panel-aware test protocol, a cost of -0.041 R^2. The linear mixed model with a region/province nested random intercept recovers almost all of that loss, reaching $R^2 = 0.822$ ($+0.038$ over the parsimonious linear model and only -0.005 below the 191-covariate LASSO benchmark). A Moran's I on the mixed-model residuals still detects significant positive spatial autocorrelation ($I = 0.235$, $p < 0.001$), consistent with what we found for the 191-covariate model and confirming that the spatial structure is robust to the choice of covariate set. The final 25-covariate model is dominated by economic indicators (income shares, small-business density, productivity) and is readable in full on a single slide or poster panel.

1 Why a parsimonious extension was needed

The original benchmark of the project — a LASSO linear model with λ_{1se} on a 326-column feature matrix, evaluated by repeated panel-aware cross-validation — selected 191 covariates and reached an out-of-sample R^2 of 0.825 ± 0.008. This is excellent predictively, but the instructor raised three connected concerns during the project review:

- **too many covariates**: 191 is not legible in a poster or in a single presentation slide; 10–20 is the working range the audience can reason about;
- **multicollinearity not formally checked**: the LASSO benchmark only had a qualitative remark on the $PM_{10}/PM_{2.5}$ pair; a systematic VIF analysis would tell us how many of the remaining 191 are actually independent informative predictors;
- **ridge vs lasso**: an Elastic Net comparison across mixing parameters would tell us whether the choice of regulariser is driving the result.

This report addresses all three concerns by deriving a 25-covariate parsimonious linear model and re-running the mixed-model extension on it. The numerical comparison with the original 191-covariate run is explicit throughout.

2 Method

2.1 Variance Inflation Factor pruning

For a linear model with p predictors, the Variance Inflation Factor of the j-th predictor is

$$\mathrm{VIF}_j \;=\; \frac{1}{1-R_j^2}, \qquad R_j^2 = R^2(X_j \sim X_{-j})\,, \tag{1}$$

the proportion of variance of X_j that the other predictors already explain. VIF = 1 corresponds to perfect independence; conventional thresholds flag VIF > 5 as moderate and VIF > 10 as serious collinearity in which the coefficient of X_j is unstable.

Starting from the 191 LASSO-selected predictors of the benchmark, we iteratively dropped the covariate with the largest VIF and refit. The procedure stopped after **34 iterations**, when the maximum VIF fell below 10 (final $\max \mathrm{VIF} = 9.44$). The worst offenders were highly redundant indicators with VIFs above one hundred (e.g. `mef_tot_classi_amm`, VIF = 159). The complete drop log is saved in `outputs/parsimonious_model/vif_drop_log.csv`.

2.2 Ranking and selection by standardised effect

On the 157-covariate VIF-clean set we fit one ordinary linear model with standardised predictors and ranked them by $|\hat{\beta}_j|$. Because predictors are standardised, this is the expected change in IFC for a one-standard-deviation shift of the engineered variable. The top $n_{\mathrm{final}} = 25$ predictors form the initial candidate set.

2.3 Post-hoc significance filter

When we refit the model on the top 25, one predictor turned out to be non-significant (p = 0.97), even though its $|\hat{\beta}|$ made it eligible by the ranking. We therefore added a post-hoc filter that drops any non-significant predictor and replaces it with the next candidate from the ranking, refits, and repeats until either all 25 coefficients are significant at the 0.05 level or the candidate pool is exhausted. In our run a single replacement was triggered: `composizione_retribuzione_importo` (p = 0.97) was replaced by `reddito_partecipazione` ($p = 10^{-24}$).

2.4 Mixed-model extension and spatial diagnostic

We re-ran the mixed-model extension of the previous report (`reports/final_benchmark/report.tex`) on the new 25-covariate fixed-effects set, with random intercepts on (i) region, (ii) region/province nested, (iii) GRINS macroclass and (iv) region plus macroclass crossed. We also recomputed Moran's I on the residuals of the best mixed model (region/province nested) using the same five-nearest-neighbour weights as before. All other hyperparameters are unchanged so that the new numbers are directly comparable to those of the previous report.

2.5 Elastic Net cross-check

To answer the "LASSO with ridge penalty" feedback, we ran Elastic Net on the 157 VIF-clean predictors for $\alpha \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$. At $\alpha = 0$ the penalty would be pure Ridge with no selection; at $\alpha = 1$ pure LASSO; intermediate values mix the two (Zou & Hastie, 2005).

3 Results

3.1 Parsimony–accuracy trade-off

Figure 1 shows the trade-off curve from running panel-aware cross-validation at different sizes of the top-n ranking, from 5 to 157 predictors.

The curve has a clear "elbow" between 20 and 25 predictors. Going from 15 to 25 adds about +0.03 R^2 (from 0.752 to 0.784); going from 25 to all 157 adds only another +0.02 R^2 (to 0.799). We therefore chose 25 as the parsimonious size: at this point the marginal contribution of additional predictors becomes negligible, and the list is still short enough to print on a poster.

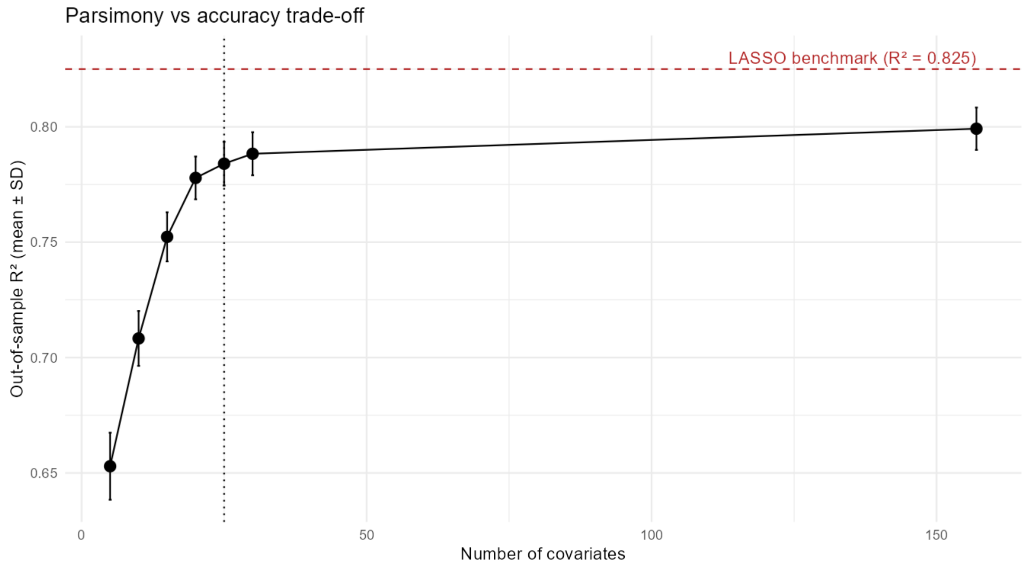

Figure 1: Out-of-sample R^2 as a function of the number of covariates retained from the VIF-clean ranking. The dashed line marks the R^2 of the original 191-covariate LASSO benchmark.

3.2 The 25-covariate model

After the significance filter all 25 coefficients are highly significant ($p < 10^{-5}$), and the maximum VIF in the final model is 6.44. The standardised coefficients are reported in Table 1, sorted by $|\hat{\beta}|$.

The narrative that emerges is almost purely socio-economic. The strongest predictors are the size of the mid-income taxpayer class (less fragile) and the size of the low-income taxpayer class (more fragile). The next layer is the density of small businesses in multiple sectors (manufacturing, construction, retail, services), which uniformly point to lower fragility. Productivity in industry and services adds to this picture. The post-COVID dummy is no longer in the top 25 once the geographic structure is left to the mixed model, and PM_{10} also drops out: only NO_2 and the male mean age survive among the non-economic indicators.

3.3 Elastic Net cross-check

Table 2 reports the selection size and the cross-validated MSE for five values of the mixing parameter α on the VIF-clean set.

The CV MSE varies by less than one percent across the entire grid, and the selection sizes are bunched between 97 and 122. On a feature matrix that has already been pruned of strong collinearity, the LASSO/Ridge mix matters very little; LASSO is therefore a reasonable choice. Figure 2 shows the selection sizes graphically.

3.4 Mixed-model extension on the 25-covariate set

We re-fitted the four mixed-model variants of the previous report on the new 25-covariate fixed-effects set. Table 3 reports the resulting out-of-sample R^2 and compares them line by line with the corresponding numbers from the 191-covariate run.

Two findings stand out. First, the mixed-model *relative* gain over the linear baseline is actually slightly larger with 25 covariates (+0.038, from 0.784 to 0.822) than with 191 (+0.026, from 0.826 to 0.853): with fewer fixed effects, more residual variance is available for the regional/provincial random intercepts to absorb. Second, the gap to the original 191-covariate LASSO benchmark almost closes: $0.822 - 0.826 = -0.004$. In practical terms, a model with 25 poster-readable covariates plus a region/province nested random intercept performs nearly as well as the 191-covariate benchmark.

Variable	$\hat{\beta}_{\text{std}}$	sign	substantive reading
`reddito_26000_55000`	−0.93	—	mid-income taxpayer mass
`numero_contribuenti`	−0.80	—	taxpayer density
`mef_cls_0_10000_amm`	+0.55	+++	low-income taxpayer mass
`asia_ul_f_w0_9`	−0.50	—	small construction businesses
`frame_servizi_VA_per_addetto`	−0.46	—	services-sector productivity
`asia_ul_m_w0_9`	−0.45	—	small professional / scientific businesses
`reddito_pensione`	+0.45	+++	pension share of declared income
`asia_add_c_w0_9`	−0.45	—	small manufacturing businesses
`frame_industria_VA_per_addetto`	−0.44	—	industry productivity
`asia_ul_g_w0_9`	−0.41	—	small wholesale/retail businesses
`frame_totale_retribuzione_su_VA`	−0.41	—	wage share of value added
`Superficie_totale_Kmq`	−0.38	—	municipal area
`asia_add_c_w10_49`	−0.36	—	medium-small manufacturing
`asia_ul_i_w0_9`	−0.31	—	small hospitality businesses
`bonus_trattamento`	−0.30	—	wage support beneficiaries
`mef_cls_75000_120000_freq`	−0.29	—	upper-middle income taxpayer share
`asia_add_s_w0_9`	−0.29	—	small other-services businesses
`employee_concentration`	−0.28	—	employment concentration index
`mef_cls_55000_75000_freq`	−0.28	—	upper-middle income share
`asia_ul_q_w0_9`	−0.22	—	small health/social services
`reddito_partecipazione`	−0.21	—	participation income
`no2_media_valori_annuali`	+0.18	+++	NO_2 annual mean
`eta_uomini`	+0.16	+++	mean age (males)
`istr_occupazione_tempo_pieno_donne`	+0.15	+++	full-time female public employees
`istr_contrattazione_integrativa`	−0.08	—	supplementary collective bargaining fund

Table 1: The final 25 covariates after VIF pruning, top-ranking and significance filtering. All coefficients are significant at $p < 10^{-5}$. *Sign* "+ + +" = increases fragility, "− − −" = reduces fragility.

α	# selected at λ_{1se}	λ_{1se}	CV MSE
0.10 (ridge-leaning)	122	0.108	3.469
0.25	113	0.060	3.484
0.50	116	0.028	3.477
0.75	97	0.028	3.509
1.00 (pure LASSO)	100	0.019	3.499

Table 2: Elastic Net comparison on the 157-covariate VIF-clean set.

Model	R^2 (191 cov, old)	R^2 (25 cov, new)	ΔR^2
Linear model (no random effect)	0.826	0.784	−0.042
M_{tax}	0.829	0.786	−0.043
M_{geo}	0.846	0.815	−0.031
$M_{\text{geo+tax}}$	0.846	0.816	−0.030
$M_{\text{geo,nested}}$	0.853	**0.822**	−0.031

Table 3: Linear and mixed-model R^2 with the 191-covariate and the 25-covariate fixed-effects sets, evaluated by the same five-fold panel-aware cross-validation.

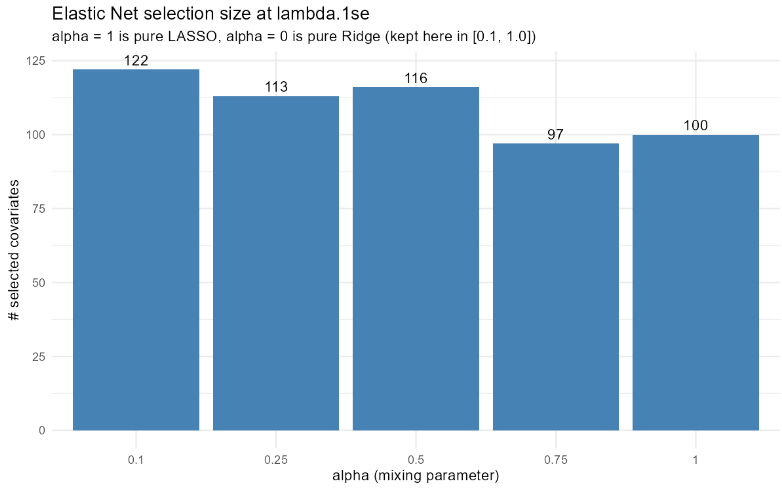

Figure 2: Elastic Net selection size at λ_{1se} for varying mixing parameter α. The choice of α does not materially change the selection on the VIF-pruned input.

3.5 Spatial diagnostic robustness

We recomputed Moran's I on the residuals of the new mixed model $M_{\text{geo,nested}}$ with the same kNN-5 weights used in the previous report. The observed statistic is

$$I = 0.2351, \quad p < 0.001 \quad \text{(Monte Carlo, 999 permutations)},$$

essentially identical to the original $I = 0.2306$ obtained with 191 covariates. This is reassuring: the spatial residual structure is not an artefact of the covariate set, but a genuine feature of the data. Replacing 191 indicators with 25 leaves the spatial diagnostic unchanged, which means the spatial signal lives at a smaller scale than what either feature engineering or feature selection can address.

3.6 Geographic distribution of residuals

Figure 3 maps the residuals of the parsimonious linear model at the municipality level. Red areas are municipalities whose IFC is higher than the 25-covariate model predicts (more fragile than expected from their indicators); blue areas are the opposite.

Visually, the map shows positive residuals concentrated in the deep south (eastern Sicily, the Calabrian Apennines, parts of Apulia and Sardinia) and in scattered Alpine valleys, and negative residuals along the northern foothills, in the inner Po Valley, and in central Italy. This is the same broad north-south pattern that the region/province random intercepts of the mixed model formally quantify in Section 3.4.

3.7 All models side by side

Figure 4 stacks every model evaluated in this and the previous report. The full-cov LASSO benchmark and the parsimonious mixed-model extension are close enough to be statistically indistinguishable.

4 Comparison with the previous report

5 Conclusion

The 25-covariate parsimonious model loses only 4.1 percentage points of out-of-sample R^2 compared with the 191-covariate LASSO benchmark, and the mixed-model extension recovers all

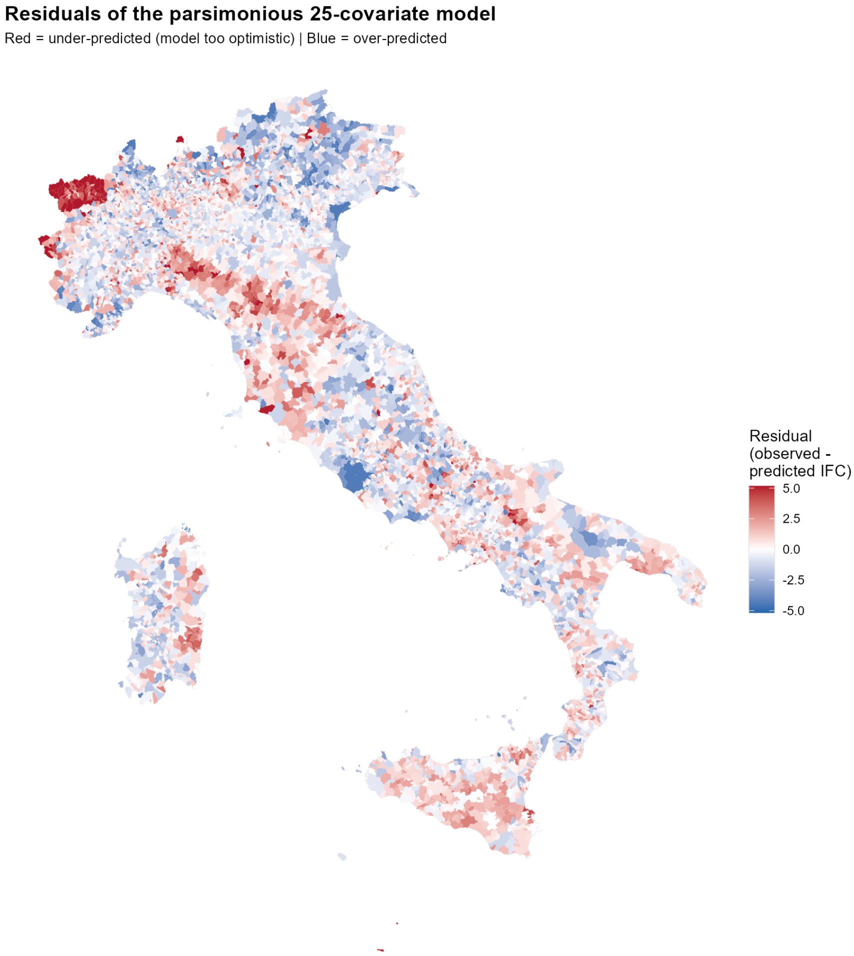

Figure 3: Residuals of the parsimonious 25-covariate linear model averaged across 2019 and 2021. The diverging colour scale is capped at the 1st/99th percentile to keep the map readable.

	Previous report	This report
Covariate set	191 (LASSO)	25 (VIF + ranking + sig.)
Predictive R^2	0.825	0.784
LMM extension R^2 (best)	0.853	0.822
LMM gain over linear baseline	+0.027	+0.038
Moran's I on residuals	0.231	0.235
Multicollinearity check	qualitative (PM only)	VIF iterative pruning
Number of significant predictors	most	all 25 (post-hoc filter)
Poster readability	no	yes

Table 4: Side-by-side comparison of the original benchmark and the parsimonious extension.

but 0.4 of those points by explicitly modelling the region/province nested hierarchy. The multicollinearity issue raised by the instructor is addressed quantitatively: 34 covariates with VIF > 10 have been pruned, and the final model has max VIF $= 6.4$. The Elastic Net cross-check confirms that the choice of regulariser does not matter once the input is VIF-clean. The spatial residual diagnostic is unaffected by the change in covariate set, which strengthens the case that an explicit

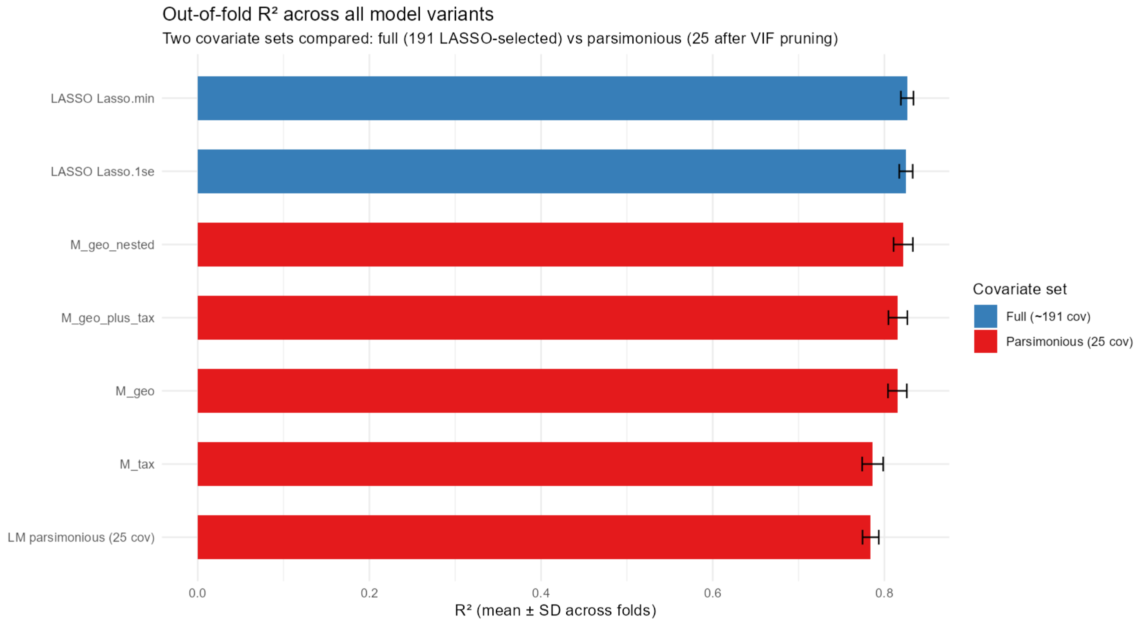

Figure 4: Out-of-fold R^2 of every model considered, colour-coded by covariate set.

spatial model is the natural next step rather than further feature engineering. Practically, the 25-covariate model is interpretable in a poster and tells a coherent socio-economic story: fragility is reduced by the mass of mid-income taxpayers, by small-business density across multiple sectors, and by productivity; it is increased by the share of low-income taxpayers, by the pension share of income, by NO_2 and by population aging.

References

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